

Declaration

I, May Xu, declare that this assignment, titled A1.1: Design Report, is my own original work and has not been copied from any other source except where explicitly acknowledged. I have not engaged in plagiarism, collusion, or any other form of academic misconduct in the preparation and submission of this assignment. All sources of information and data used in this assignment have been properly cited and referenced in accordance with the prescribed guidelines. I have not used unauthorized assistance in the preparation of this assignment and have not allowed any other student to copy my work. I am aware that any breach of academic integrity may result in disciplinary action as per the [policies of Monash University](#), which may include failing this assignment or the course, and further academic penalties.

Signature: May Xu

Date: 14/08/2026

Github Check

Enter your Github details here.

Github Username <i>Enter your username here</i>	szzy-9
Repository Shared? <i>Have you started and shared your assignment repository with your tutor yet?</i>	Yes

Self-Evaluation

Rate your performance for each criteria. Put a ☒ (tick) in the box where you think your work belongs.

Criteria	Exceeds Expectations	Meets Expectations	Needs Improvement	Fail to meet expectations
Executive Summary	☒			
User Personas	☒			
Sitemap	☒			
Wireframes/Low-fidelity Prototype	☒			

Section 1: Executive Summary

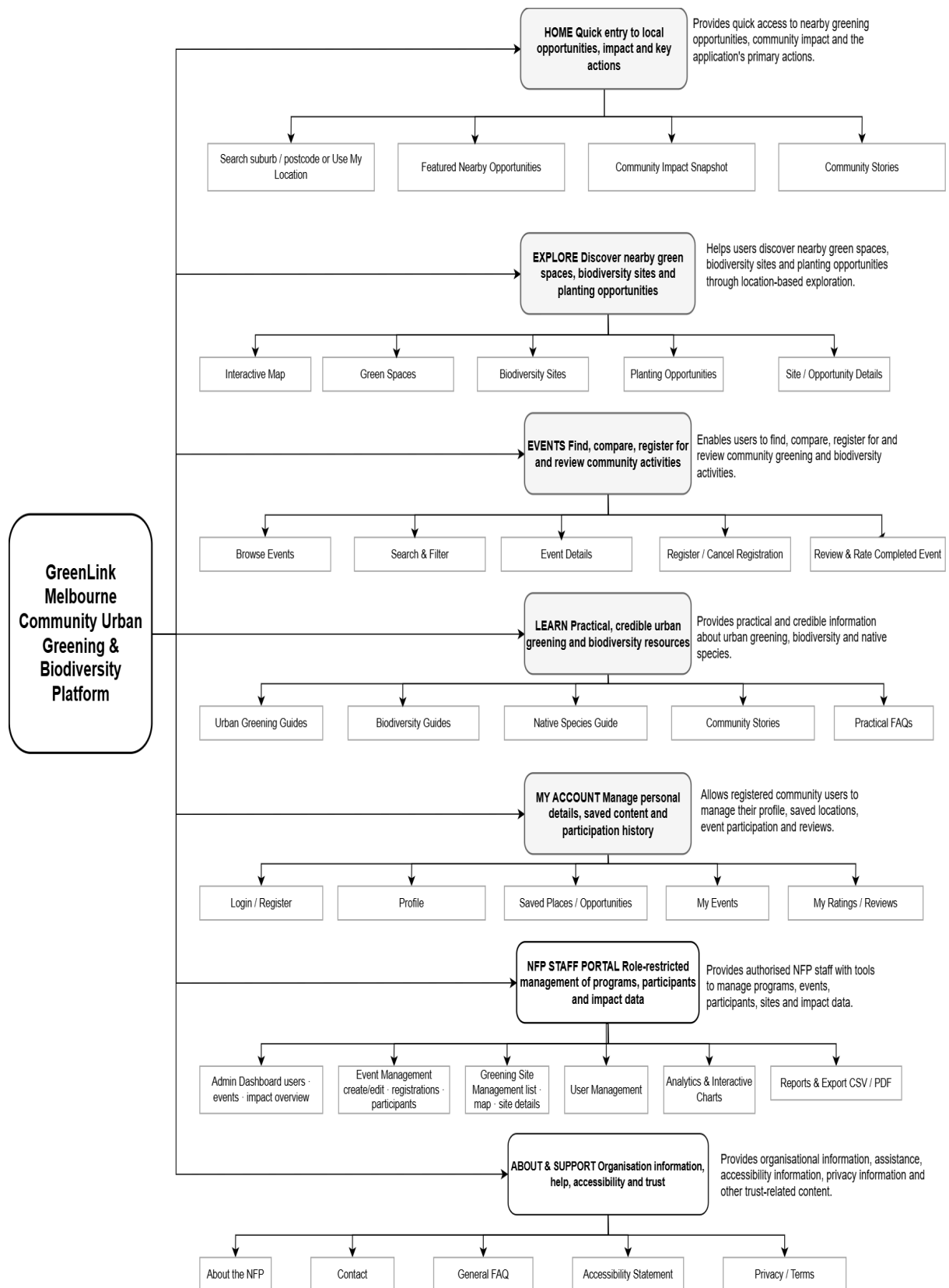
GreenLink Melbourne is a proposed web application for a Melbourne-based not-for-profit organisation working to strengthen urban greening, biodiversity and community climate resilience, particularly for underrepresented communities. The client's objectives are to expand community reach, improve access to environmental programs and information, and streamline the management of community activities. Recent research reinforces the need for this work: urban biodiversity can support climate adaptation, human wellbeing and environmental justice, while unequal access to green infrastructure can leave some communities with fewer benefits (Kowarik et al., 2025; Li et al., 2024). In Victoria, recent evidence also shows lower tree-canopy coverage in more socioeconomically disadvantaged school areas, with particularly low coverage in Melbourne's western suburbs (Abel et al., 2026).

The application will serve three primary user groups: local residents seeking accessible information about nearby green spaces and biodiversity; community volunteers looking for practical opportunities to participate in planting and restoration activities; and NFP staff responsible for coordinating events, participants and greening programs. This multi-user structure supports both public engagement and internal program management within one platform.

Original design research was conducted across the websites of Greening Australia, Landcare Australia, CERES and Greenfleet. Landcare Australia demonstrates the value of location-based discovery through postcode, suburb and geolocation search; CERES foregrounds events and hands-on community experiences; Greening Australia combines science-based resources with audience-specific pathways; and Greenfleet uses concise impact metrics to communicate outcomes. These patterns informed an action-oriented rather than brochure-style design.

Accordingly, GreenLink Melbourne has three primary goals. First, it will help residents discover nearby green spaces, biodiversity sites and community activities through an interactive map and location-based search. Second, it will make participation easier by enabling users to browse, filter, register for and review environmental events. Third, it will provide NFP staff with a role-based administration portal for managing events, participants, sites and impact data. Supporting resources, accessible interface design and transparent impact reporting will connect environmental knowledge with practical local action. Overall, the application is intended to make urban greening easier to discover, easier to participate in and easier for the NFP to coordinate.

Section 2: Sitemap



Section 3: User Personas

Persona 1 Maya Chen: Local Resident

Demographics

Maya is 31 years old and lives in Footscray, Melbourne. She works full-time as an administrative officer and usually accesses websites on her phone. She has moderate digital confidence and is interested in environmental issues, but she is not an expert in biodiversity or urban planning.

Goals and Needs

Maya wants to understand what green spaces, biodiversity sites and community greening projects are available near her home. She would like clear information about local vegetation and environmental activities without having to read technical reports. She also wants to save interesting locations and quickly understand whether a place or activity is relevant and easy to reach.

Pain Points

Information about urban greening is often spread across council pages, environmental organisations and separate event websites. Maya finds long, text-heavy pages difficult to scan and may be unsure which information is current or relevant to her suburb. She is unlikely to participate if she cannot quickly understand where an activity is located, what it involves or whether it is suitable for beginners.

How GreenLink Addresses Her Requirements

GreenLink will provide a location-based Explore section with an interactive map, suburb/postcode search and clear site cards. Each site or opportunity page will summarise its purpose, biodiversity value, location and related activities in plain English. Maya can save places through My Account and access short learning guides explaining urban greening and native biodiversity. A responsive, visually clear interface will support quick mobile use and reduce the effort required to find reliable local information.

 MAYA • 31 years old • Footscray • Local resident • Mobile user "I want an easy way to see what green spaces and environmental activities are actually near me."	DESCRIPTION Maya is a busy local resident who cares about the environment but has limited knowledge of urban biodiversity. She mainly uses her phone and wants a simple way to discover nearby green spaces, community projects, and activities without searching across multiple websites.	PERSONAL CHARACTERISTICS • Environmentally aware • Practical • Curious • Time-conscious • Moderately tech-confident	HOBBIES AND INTERESTS • Walking in local parks • Gardening • Weekend markets • Exploring Melbourne • Community activities
	USER ROLE • Local resident • Information seeker • Potential participant	GOALS • Find nearby green spaces • Discover local greening activities • Understand biodiversity information • Save useful places and opportunities	CHALLENGES • Information spread across multiple websites • Long and technical environmental content • Difficulty identifying nearby opportunities • Unclear or outdated activity information
	USER'S JOURNEY Discover → Explore → Participate → Share	NEEDS • Location-based search • Interactive map • Clear, concise information • Mobile-friendly access • Saved places and activities	SOURCES OF INFO • Council websites • Google search • Social media • Community groups • Friends and word of mouth

Persona 2 Liam O'Connor: Community Volunteer

Demographics

Liam is 23 years old, studies at university and lives in Clayton. He has a limited budget, does not own a car and mainly travels by public transport, bicycle or walking. He is highly comfortable with digital services and regularly uses event-booking and map applications.

Goals and Needs

Liam wants to take practical climate action rather than only read about environmental problems. He is looking for weekend tree-planting, habitat-restoration and biodiversity activities that are reasonably close to him. Before registering, he needs to compare dates, locations, activity types, accessibility, available places and required experience. He also wants to keep track of upcoming activities and receive clear confirmation after booking.

Pain Points

Liam often finds volunteering opportunities through separate social-media posts, council pages and organisation websites. Event information can be inconsistent, outdated or missing important details such as capacity, transport access and what participants should bring. Re-entering similar information for different activities is inconvenient, and he has no central place to review his participation history.

How GreenLink Addresses His Requirements

The Events section will allow Liam to browse and filter opportunities by date, location and activity type, while the interactive map will show nearby options. Event detail pages will display capacity, accessibility, requirements and organiser information before registration. After logging in, Liam can register for activities, view My Events, manage permitted cancellations and receive confirmation information. After completing an event, he can submit a rating and review. These features create a clearer pathway from environmental interest to practical community participation.



LIAM

- 23 years old
- Clayton
- University student
- Community volunteer

"I want to find practical climate action near me, not search five separate websites."

ABOUT THE USER

Liam wants to take practical climate action rather than only read about environmental problems. He uses his phone and prefers clear, map-based information about nearby weekend activities.

PROBLEMS

- Event information is scattered across websites and social posts
- Transport, accessibility and capacity details are unclear
- Hard to compare date, distance and activity type

CHALLENGES

Repeated registration forms waste time, and Liam has no central participation history. If event details look outdated or incomplete, he is less likely to attend.

DIGITAL BEHAVIOUR

- Mobile-first
- Map-savvy
- Uses booking apps
- Comfortable with filters
- Expects quick confirmation

GOALS AND NEEDS

- Find nearby weekend planting or biodiversity activities
- Filter by location, date and activity type
- Register quickly and track events in one place

Persona 3 Priya Nair: NFP Program Coordinator

Demographics

Priya is 42 years old and works full-time as a program coordinator for a Melbourne environmental not-for-profit organisation. She manages community greening activities, volunteers and project information. She is confident with common office software but does not have advanced technical or data-analysis skills. She mainly uses a desktop computer during work.

Goals and Needs

Priya needs one system for creating events, managing registrations, maintaining greening-site information and monitoring community participation. She wants to know which events are full, how many people are participating, which sites are active and what outcomes can be reported to managers, partners or funders. She also needs reliable export functions for reports and participant information.

Pain Points

Priya currently depends on spreadsheets, email threads and separate documents. This creates duplicated work and makes it difficult to identify the latest information. Manual participant tracking is time-consuming, while changes to an event may not be communicated consistently. Preparing basic impact reports can also require combining information from several sources.

How GreenLink Addresses Her Requirements

The role-restricted NFP Staff Portal will provide an admin dashboard, event management, greening-site management, user management, analytics and export tools. Priya can create or update public events, monitor registrations, review participant tables and manage site information displayed on the community map. Dashboard summaries and interactive charts will provide quick visibility of participation and program activity, while CSV/PDF export will support reporting. This integrated workflow reduces administrative duplication and connects internal NFP operations directly with the public-facing community experience.

 PRIYA • 42 years old • Melbourne NFP • Program Coordinator • Staff/Admin user "MSGR is an absolute must-have for any team looking to maximize productivity!"	DESCRIPTION Priya coordinates community greening programs for a Melbourne NFP. She manages events, volunteers and site information, but currently relies on spreadsheets, emails and separate documents.	PERSONALITY • Organised • Community-focused • Practical • Data-conscious • Time-poor	WORK CONTEXT • Community events • Urban greening programs • Volunteer coordination • Council/NFP reporting • Impact measurement
	USER ROLE NFP staff. She can create, edit and monitor public-facing events and greening-site information.	GOALS • Manage events and registrations • Monitor site and participant data • Prepare reports for partners and funders	PAIN POINTS • Fragmented spreadsheets and emails • Duplicate participant tracking • Latest event data is hard to confirm
	USER'S JOURNEY Create → Manage → Monitor → Report	NEEDS • Role-based staff portal • Interactive tables and dashboard • CSV/PDF export for reporting	GREENLINK RESPONSE Event and site management, analytics, charts and export tools connect internal NFP operations with the public map and event pages.

Section 4: Wireframes/Low-fidelity prototype

BR E.2 — Geo Location: Explore Nearby Greening Opportunities

Location search → map results → selected opportunity. Planned map features: location-based discovery and route.

GreenLink Melbourne Home Explore Events Learn My Account

Explore urban nature near you

Search suburb or postcode

Use My Location

Type: All ▼ Distance: 5 km ▼

INTERACTIVE MAP

○ Green space

△ Biodiversity site

● Planting event

Riverside Planting Day Sat 22 Aug · 1.8 km Beginner friendly 12 places remaining

View Details

BR E.2 — Geo Location: Opportunity Details and Directions

Selected opportunity → details → lightweight route information, demonstrating a second non-trivial geolocation feature.

GreenLink Melbourne Explore Events Learn

Riverside Planting Day

Sat 22 Aug · 10:00–13:00 Footsray · 1.8 km away Beginner friendly · Accessible toilets

MAP PREVIEW ● Event location

About this activity Community planting of native shrubs to improve habitat connectivity. What to bring Closed shoes · water · sun protection

Get Directions

Register

Save

GreenLink Melbourne Directions

Directions to Riverside Planting Day

From: Current location ▼

To: Riverside planting site

ROUTE MAP Start ○ | ● Destination Walking route highlighted

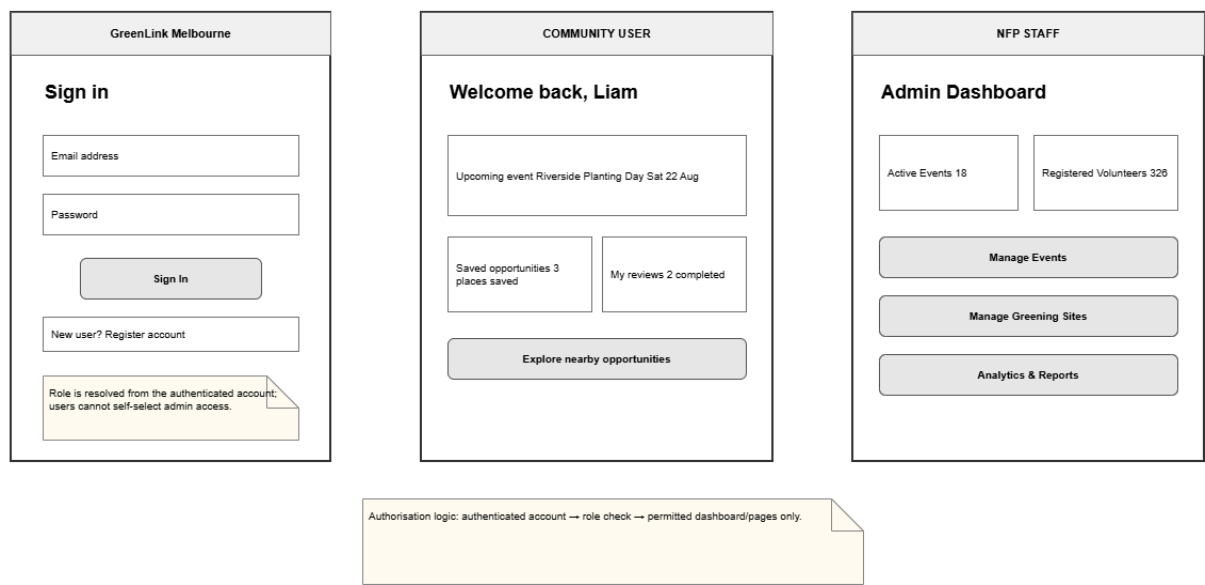
Walking · 22 min · 1.7 km

Open directions

Design rationale: navigation remains lightweight so GreenLink supports participation without becoming a full transport-routing product.

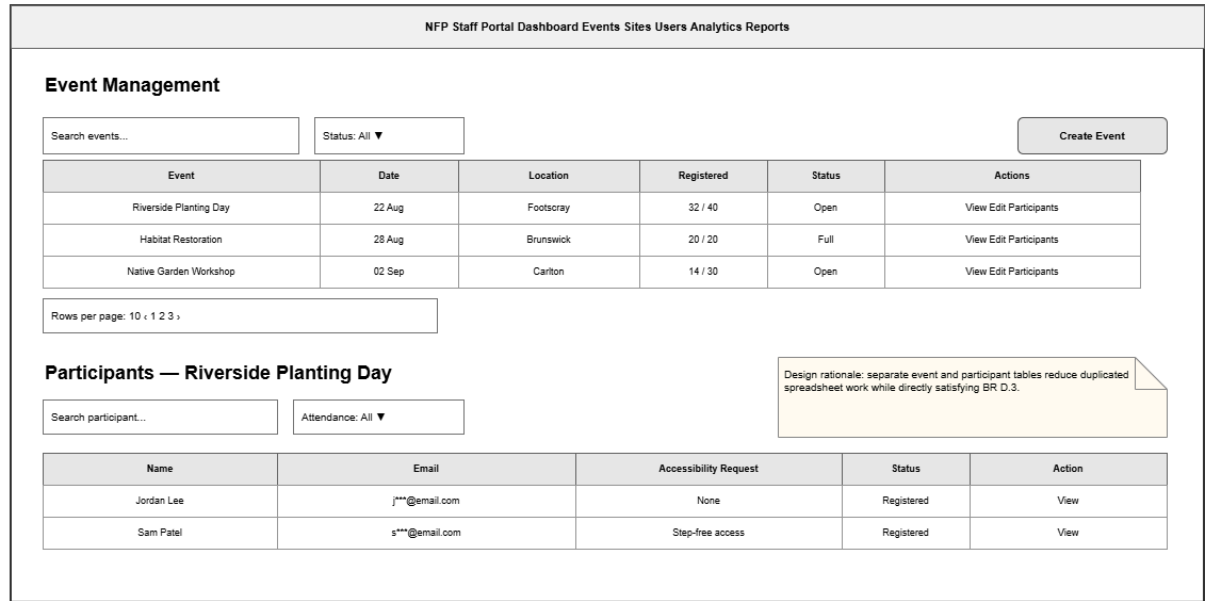
BR C.2 — Role-Based Authentication: Community User vs NFP Staff

The same sign-in entry leads to different authorised interfaces and functions.



BR D.3 — Interactive Table Data: NFP Event and Participant Management

Two admin tables support search, column filtering, sorting and pagination with a maximum of 10 rows per page.



Declaration: Additional Help

Any tools that you used (including Gen AI or existing code reuse) must be declared here.

Note: GenAI is not allowed for coding purposes in any assignment,

However, you may use GenAI for brainstorming, problem solving and learning. You need to declare all such uses here. One row per help used. More details on how to acknowledge the use of Gen AI can be found [here](#).

Name	Description
Gemini for brainstorming relevant websites	I used Gemini to brainstorm relevant not-for-profit and environmental organisation websites that could be analysed for design inspiration.
Gemini for literature search	I used Gemini to help identify recent academic literature related to urban greening, biodiversity, tree canopy equity, and community participation. I then reviewed the relevant sources and used them to support the background research for the project.

Reference

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